**MS:**

4.3 $f(x) = x + x^3 \implies f(3)$ is defined: $f(3) = \boxed{30}$

4.4 $f(4)$ is defined: $f(4) = \boxed{9}$

4.8 $f(x) = \frac{\ln(x)}{x}$

Yes, discontinuity at $x = 0$ which is not in $[2, \infty)$

4.9 $f(x) = \begin{cases} x^3 - 3x + 4 & : x \leq 3 \\ x^2 & : x > 3 \end{cases}$

i) $f(3) = \boxed{22}$

ii) $\lim_{x \to 3^-} f(x) = \boxed{22} \neq \lim_{x \to 3^+} f(x) = \boxed{9}$

$\implies f(x)$ can be made continuous by adding 13 to $x^2$

5.2 (a) $\lim_{h \to 0} \frac{6-6}{h} = \boxed{0}$

(b) $\lim_{h \to 0} \frac{3(x+h)^2 - 3x^2}{h}$

$$= \lim_{h \to 0} \frac{3(x^2 + 2xh + h^2) - 3x^2}{h} = \lim_{h \to 0} \frac{3x^2 + 6xh + 3h^2 - 3x^2}{h}$$

$$= \lim_{h \to 0} \frac{6xh + 3h^2}{h} = \lim_{h \to 0} \frac{h(6x + 3h)}{h} = \lim_{h \to 0} (6x + 3h) = \boxed{6x}$$

(c) $\lim_{h \to 0} \frac{(x+h)^3 - 2(x+h)^2 - 1 - (x^3 - 2x^2 - 1)}{h}$

$$= \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - 2(x^2 + 2xh + h^2) - 1 - x^3 + 2x^2 + 1}{h}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - 4xh - 2h^2}{h}$$

$$= \lim_{h \to 0} \frac{h(3x^2 + 3xh + h^2 - 4x - 2h)}{h} = \lim_{h \to 0} (3x^2 + 3xh + h^2 - 4x - 2h) = \boxed{3x^2 - 4x}$$

(h) $\lim_{h \to 0} \frac{5(x+h)^5 + 4(x+h)^4 + 3(x+h)^3 + 2(x+h)^2 + (x+h) + 1) - (5x^5 + 4x^4 + 3x^3 + 2x^2 + x + 1)}{h}$

$$= \lim_{h \to 0} \frac{5(x^5 + 5x^4h + 10x^3h^2 + 10x^2h^3 + 5xh^4 + h^5) + 4(x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4)}{h}$$

$$+\frac{3(x^3+3x^2h+3xh^2+h^3)+2(x^2+2xh+h^2)+(x+h)+1-(5x^5+4x^4+3x^3+2x^2+x+1)}{h}$$

$$=\lim_{h\to 0}\frac{5x^5+25x^4h+50x^3h^2+50x^2h^3+25xh^4+5h^5+4x^4+16x^3h+24x^2h^2+16xh^3+4h^4}{h}$$

$$+\frac{3x^3+9x^2h+9xh^2+3h^3+2x^2+4xh+2h^2+x+h+1-(5x^5+4x^4+3x^3+2x^2+x+1)}{h}$$

$$=\lim_{h\to 0}\frac{25x^4h+50x^3h^2+50x^2h^3+25xh^4+5h^5+16x^3h+24x^2h^2+16xh^3+4h^4+9x^2h+9xh^2+3h^3}{h}$$

$$+\frac{4xh+2h^2+h}{h}$$

$$=\lim_{h\to 0}\left(25x^4+50x^3h+50x^2h^2+25xh^3+5h^4+16x^3+24x^2h+16xh^2+4h^3+9x^2+9xh+3h^2+4x+2h+1\right)$$

$$=\boxed{25x^4+16x^3+9x^2+4x+1}$$

6.1

(a) $0$

(b) $6x$

(c) $3x^2-4x$

(d) $4x^3+5$

(e) $8x^7$

(f) $-3x^{-4}$

(g) $3ax^2$

(h) $6x^{-\frac{1}{2}}$

(i) $nax^{n-1}$

(j) $3-12x^2$

(k) $(13+6x^2)(x^5-4x+r)+(5x^4-4)(13x+2x^3)$

(l) $3(x-3)^2$

(m) $\frac{2x(x+1)-x^2-1}{(x+1)^2}$

(n) $35x^6+28x^3+6x$

(o) $40x^7+70x^6-30x^5-25x^4+12x^3+21x^2-4x+1$

(p) $3x^2+2x$

(q) $4x^3-3x^2+2x-1$

(r) $6x(2x^3+3x+5)+(6x^2+3)(3x^2+4)$

(s) $(15x^2+8x+3)(7x^5+6x^4+5x^3+4x^2)$

$$+(35x^4+24x^3+15x^2+8x)(5x^3+4x^2+3x+2)$$

(t) $6x+6x^2+3$

(u) $25x^4+9x^2+1-16x^3-4x$

(v) $2(x+5)$

(w) $2(2x+1)(x^2+x+2)$

(x) $2\left(\frac{2x(x+1)-x^2-1}{(x+1)^2}\right)\left(\frac{x^2+1}{x+1}\right)$

(y) $\frac{(3x^2+2x+1)(x^2+x+1)-(2x+1)(x^3+x^2+x+1)}{(x^2+x+1)^2}$

6.2 (b) $f(x) = (x^3+2)\ln(x^4-5x+3)$

$$f'(x) = 3x^2\ln(x^4-5x+3) + (x^3+2)\left(\frac{4x^3-5}{x^4-5x+3}\right)$$

(d) $f'(x) = \left(1-\frac{1}{x}\right)e^{x-\ln x+5}$

(f) For the derivative of $a^x$ use the identity: $a^x = e^{x\ln a}$. Then apply the chain rule: $\frac{d}{dx}e^{x\ln a} = e^{x\ln a}\cdot\ln a = a^x\cdot\ln a$.

$\rightarrow$ Now find $f'(x)$.

$$f'(x) = a^x\ln(a)x^2 + 2a^x x - 2x\ln(b)b^{x^2}$$

(g) $f'(x) = 5e^{5x}$

(k) $f'(x) = \frac{1}{x}e^{\ln(2x)} = 2$

(l) $f'(x) = g'(x)e^{g(x)}$ , where $g'(x) = 21x^2+10x-3+\frac{1}{x}$

6.3 Begin by noting that $a^{\log_a(x)} = x$, since the two are inverse functions of each other.

Now differentiate both sides with respect to $x$, using the chain rule on the left-hand side. The right-hand side has a derivative of 1.

To compute the left-hand side, let $g(x) = a^x$ and $h(x) = \log_a(x)$. Then:

$$g'(x) = \ln(a)\cdot a^x \quad \text{and} \quad h'(x) = \frac{d}{dx}\log_a(x)$$

The chain rule tells us that the derivative of the composition $g(h(x)) = a^{\log_a(x)}$ is:

$$\frac{d}{dx}\left(a^{\log_a(x)}\right) = \ln(a)\cdot a^{\log_a(x)}\cdot h'(x)$$

Since $a^{\log_a(x)} = x$, we substitute to get:

$$\ln(a)\cdot x\cdot h'(x) = 1$$

Solving for $h'(x)$, we find:

$$h'(x) = \frac{1}{x\ln(a)}$$

Therefore, we have shown that: $\frac{d}{dx}\log_a(x) = \frac{1}{x\ln(a)}$

Cont.

## From Sydsaeter and Hammond

6.3 (a) $y' = 2, \quad y'' = 0$

(b) $y' = 3x^8, \quad y'' = 24x^7$

(c) $y' = -x^9, \quad y'' = -9x^8$

(d) $y' = 21x^6, \quad y'' = 126x^5$

(e) $y' = \frac{1}{10}, \quad y'' = 0$

(f) $y' = 5x^4 + 5x^{-6}, \quad y'' = 20x^3 - 30x^{-7}$

(g) $y' = x^3 + x^2, \quad y'' = 3x^2 + 2x$

(h) $y' = -\frac{1}{x^2} - \frac{3}{x^4}, \quad y'' = \frac{2}{x^3} + \frac{12}{x^5}$

6.5.4 (a) $\lim_{x\to 2}(x^2 + 3x - 5)$

$$= 4 + 6 - 5 = \boxed{5}$$

(b) $\lim_{y\to -3} \frac{1}{y+8} = \boxed{\frac{1}{5}}$

(c) $\lim_{x\to 0} \frac{x^3-2x-1}{x^5-x^2-1} = \lim_{x\to 0} \frac{-1}{-1} = \boxed{1}$

(d) $\lim_{x\to 0} \frac{x^3+3x^2-2x}{x} = \lim_{x\to 0} \frac{x(x^2+3x-2)}{x} = \lim_{x\to 0}(x^2 + 3x - 2) = \boxed{-2}$

(e) $\lim_{h\to 0} \frac{(x+h)^3-x^3}{h}$

$$= \lim_{h\to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h\to 0} \frac{h(3x^2 + 3xh + h^2)}{h} = \lim_{h\to 0}(3x^2+3xh+h^2) = \boxed{3x^2}$$

(f) $\lim_{x\to 0} \frac{(x+h)^3-x^3}{h} \quad (h \neq 0)$

$$= \lim_{x\to 0} \frac{h^3}{h} = \boxed{h^2}$$

7.1.1 $6x + 2\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \boxed{-3x}$

Check: solving for $y$ gives $y = \frac{5-3x^2}{2}$, and differentiating directly: $y' = -3x$. ✓

Cont.

7.1.2 $2xy + x^2\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{-2xy}{x^2} = \boxed{\frac{-2y}{x}}$

$$\implies \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{-2y}{x}\right)$$

$$= \frac{1}{x^2}(-2x\frac{dy}{dx} + 2y) = \frac{1}{x^2}(4y+2y) = \boxed{\frac{6y}{x^2}}$$

Check: solving for $y$ gives $y = x^{-2}$, so $y' = -2x^{-3} = \frac{-2y}{x}$ and $y'' = 6x^{-4} = \frac{6y}{x^2}$. ✓

**<u>From Gill</u>**

5.8 (a) $\lim_{x\to\infty} \frac{\ln(x)}{x}$

$$\lim_{x\to\infty} \frac{\ln(x)}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \lim_{x\to\infty} \frac{1}{x} = 0$$

(b) $\lim_{x\to\infty} \frac{3^x}{x^3}$

$$\lim_{x\to\infty} \frac{3^x}{x^3} = \lim_{x\to\infty} \frac{3^x \ln(3)}{3x^2} = \lim_{x\to\infty} \frac{3^x(\ln(3))^2}{6x} = \lim_{x\to\infty} \frac{3^x(\ln(3))^3}{6} = \infty$$

(c) $\lim_{y\to\infty} \frac{3e^y}{y^3}$

$$\lim_{y\to\infty} \frac{3e^y}{y^3} = \lim_{y\to\infty} \frac{3e^y}{3y^2} = \lim_{y\to\infty} \frac{e^y}{2y} = \lim_{y\to\infty} \frac{e^y}{2} = \infty$$

6.9 (a)

$$f'(x) = 20x^3 + 9x^2 - 22x + 1$$
$$f''(x) = 60x^2 + 18x - 22$$
$$f'''(x) = 120x + 18$$

(b)

$$h'(z) = 333z^2 - 121$$
$$h''(z) = 666z$$
$$h'''(z) = 666$$

(c)

$$f'(y) = \frac{1}{2}y^{-1/2} - \frac{7}{2}y^{-9/2}$$
$$f''(y) = -\frac{1}{4}y^{-3/2} + \frac{63}{4}y^{-11/2}$$
$$f'''(y) = \frac{3}{8}y^{-5/2} - \frac{693}{8}y^{-13/2}$$

**<u>Additional Questions:</u>**

Cont.

(a) Suppose $\pi(Q) = QP(Q) - cQ$, where $P$ is a differentiable function and $c$ is a constant. Find an expression for $\frac{d\pi}{dQ}$.

$$\frac{d\pi}{dQ} = P(Q) + QP'(Q) - c$$

(b) Suppose $\pi(L) = PF(L) - wL$, where $F$ is a differentiable function and $P$ and $w$ are constants. Find an expression for $\frac{d\pi}{dL}$.

$$\frac{d\pi}{dL} = PF'(L) - w$$

The End.